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Review

Status of vancomycin-resistant *Enterococcus* in species of wild birds: A systematic review and meta-analysis

Yusuf Wada ^{a,b}, Abdulhamid Bruor Ibrahim ^b, Yahaya Abdullahi Umar ^c,
Hafeez A. Afolabi ^d, Mustapha Wada ^e, Mohammed Alissa ^f,
Kawthar Amur Salim Al Amri ^g, Amani Ahmed AL Ibrahim ^h, Mona A. Al Fares ⁱ,
Hawra Albayat ^j, Mohammed Kamal AlKhathlan ^k, Nawal A. Al Kaabi ^{l,m},
Maha F. Al-Subaie ^{n,o,p}, Mubarak Alfaresi ^{q,r}, Hayam A. Alrasheed ^s, Ali A. Rabaan ^{p,t,u},
Chan Yean Yean ^{a,v,*}, Abdul Rahman Zaidah ^{a,v,*}

^a Department of Medical Microbiology and Parasitology, School of Medical Sciences, Universiti Sains Malaysia, Kubang Kerian 16150, Malaysia

^b Department of Zoology, Faculty of Life Sciences, Ahmadu Bello University, Zaria 810211, Nigeria

^c Department of Biological Sciences, Nigerian Defence Academy, Kaduna, Nigeria

^d Department of General Surgery, School of Medical Sciences, Universiti Sains Malaysia, Kubang Kerian 16150, Malaysia

^e Faculty of Veterinary Medicine, Ahmadu Bello University, Zaria, Nigeria

^f Department of Medical Laboratory Sciences, College of Applied Medical Sciences, Prince Sattam bin Abdulaziz University, Al-Kharj 11942, Saudi Arabia

^g Infection and Control Department, Armed Forces Hospital, Azaibah 130, Oman

^h Department of Pharmacy, Jubail General Hospital, Jubail 35514, Saudi Arabia

ⁱ Department of Internal Medicine, King Abdulaziz University Hospital, Jeddah 21589, Saudi Arabia

^j Infectious Disease Department, King Saud Medical City, Riyadh 7790, Saudi Arabia

^k International programs school, Ministry of education, Dammam 34236, Saudi Arabia

^l College of Medicine and Health Science, Khalifa University, Abu Dhabi 127788, United Arab Emirates

^m Sheikh Khalifa Medical City, Abu Dhabi Health Services Company (SEHA), Abu Dhabi, 51900, United Arab Emirates

ⁿ Research Center, Dr. Sulaiman Alhabib Medical Group, Riyadh 13328, Saudi Arabia

^o Department of Infectious Diseases, Dr. Sulaiman Alhabib Medical Group, Riyadh 13328, Saudi Arabia

^p College of Medicine, Alfaisal University, Riyadh, Saudi Arabia

^q Department of Microbiology, National Reference laboratory, Cleveland clinic Abu Dhabi, Abu Dhabi 92323, United Arab Emirates

^r Department of Pathology, College of Medicine, Mohammed Bin Rashid University of Medicine and Health Sciences, Dubai 505055, United Arab Emirates

^s Department of Pharmacy Practice, College of Pharmacy, Princess Nourah bint Abdulrahman University, Riyadh 11671, Saudi Arabia

^t Molecular Diagnostic Laboratory, Johns Hopkins Aramco Healthcare, Dhahran 31311, Saudi Arabia

^u Department of Public Health and Nutrition, The University of Haripur, Haripur 22610, Pakistan

^v Hospital Universiti Sains Malaysia, Universiti Sains Malaysia, Kubang Kerian 16150, Malaysia

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ABSTRACT

Wild birds could be a reservoir of medically relevant microorganisms, particularly multidrug-resistant *Enterococcus* spp. Resistant bacteria's epidemiology and transmission between animals and humans has grown, and their zoonotic potential cannot be ignored. This is the first study to evaluate the status of vancomycin resistant enterococci (VRE) in various wild bird species using meta-analysis and a systematic review. In this study, the pooled prevalence was obtained by analyzing data from published articles on the occurrence of VRE in wild bird species. It's unclear how the antibiotic resistance gene transfer cycle affects wild birds. Google Scholar and PubMed were used to conduct the research. The data and study methodology was assessed and extracted by two reviewers independently, with a third reviewing the results. Heterogeneity between study and publication bias were analyzed using the random effect model. Thirty-eight studies were included in the meta-analysis. 382 out of the 4144 isolates tested, were VRE. The pooled prevalence of VRE among wild birds was estimated at 11.0% (95% CI; 6.9–17.2%; $I^2 = 93.204\%$; $P < 0.001$).

* Corresponding authors at: Department of Medical Microbiology and Parasitology, School of Medical Sciences, Universiti Sains Malaysia, Kubang Kerian 16150, Malaysia.

E-mail addresses: yychan@usm.my (C.Y. Yean), drzaidah@usm.my (A.R. Zaidah).

There was high variability between study ($t^2 = 2.156$; heterogeneity $I^2 = 93.204\%$ with chi-square (Q) = 544.413, degrees of freedom (df) = 37, and $P < 0.001$). Egger's test verified the funnel plot's bias, while result from the leave-one-out forest plot had no effect on the pooled prevalence.

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Introduction

In addition to being a bacteria found in the gut microbiome of both humans and animals, enterococci are also known to be nosocomial [1–3]. *Enterococcus faecalis* and *Enterococcus faecium* were previously considered to be among some of the most prevalent bacteria associated in infections acquired in hospital [4,5]. The occurrence of enterococcal-resistant strains was considered to be as a result of human antibiotic usage and their utilization of avoparcin as feed additives in the animal husbandry to stimulate animal growth [6–9].

Vancomycin is an antibiotic used to typically treat Gram-positive bacterial infections and has also been used for human prophylaxis. As a result, the large upsurge in the frequency of vancomycin resistant enterococci (VRE) both in and around hospitals is seen as a global health issue [10–12]. Medical professionals are facing an increasing number of clinical challenges related to antimicrobial resistance. Specifically, managing VRE in the hospital setting has become more difficult. Enterococci are gram-positive, facultatively anaerobic cocci that live in pairs or chains in the gastrointestinal (GI) tract and typically interact with humans in a commensal manner. They can, however, result in a number of infections; the most frequent ones being endocarditis, bacteremia, urinary tract infections (UTI), and intraabdominal infections. They may infrequently result in pneumonia, septic arthritis, osteomyelitis, or meningitis. It is additionally more challenging to decide when and how to treat these infections because vancomycin-resistant *Enterococcus* frequently appears as a colonising bacterium that does not always contribute to infection. Despite these challenges, it has been demonstrated that infection with VRE increases cost and mortality.

The influence of the antibiotic resistance gene transfer cycle in wild species of birds or mammals is unclear. Wild birds, according to studies several studies, could be a reservoir of therapeutically relevant microorganisms, particularly multidrug-resistant *Enterococcus* spp [13–15]. Due to the varieties of foods they eat, their mode of life, and the complexity of their many ecosystems, wild birds are considered an essential

marker for tracking the development of antibiotic-resistant bacteria. Infectious diseases like West Nile virus, bird flu, and harmful bacteria like *Salmonella* have been detected in birds. Accordingly, songbirds for instance can carry a wide range of germs, all of which are hazardous to birds [16].

Consequently, antibiotic resistance in various species of birds is a grave public health issue that requires a subtle but effective response. Resistant bacteria's epidemiology and transmission between animals and humans has grown, and their zoonotic potential cannot be ignored.

This study was intended to assess the status of VRE in wild bird species. This could offer background information for awareness and the development of suitable and focused measures to decrease antibiotic resistance in wild bird species around the world. A meta-analysis would enable us to present a metric that is precise and trustworthy while also validating the findings of numerous research reporting VRE in wild birds.

Materials and methods

Study design and protocol

The checklist for this investigation was based on the Preferred Reporting Items for Systematic Reviews and Meta-Analysis Protocol (PRISMA-P 2020) criteria [17] (Supplementary File S1). This study was registered on PROSPERO with registration ID CRD42022304440.

Literature review

To ensure that there is not any meta-analysis on the occurrence of VRE in wild bird, we searched the PROSPERO and the database of abstracts of reviews of effects (DARE). After that, Google Scholar and PubMed were assessed by reviewers for published articles on the occurrence of VRE in wild bird species. PubMed database was searched using the search strategy; ("vancomycin resistant enterococci"[MeSH Terms] OR ("vancomycin resistant"[All Fields] AND "enterococci"[All Fields]) OR "vancomycin resistant enterococci"[All Fields] OR

("vancomycin"[All Fields] AND "resistant"[All Fields] AND "enterococcus"[All Fields]) OR "vancomycin resistant enterococcus"[All Fields] OR ("vancomycin resistant enterococci"[MeSH Terms] OR ("vancomycin resistant"[All Fields] AND "enterococci"[All Fields]) OR "vancomycin resistant enterococci"[All Fields] OR ("vancomycin"[All Fields] AND "resistant"[All Fields] AND "enterococci"[All Fields]) OR "vancomycin resistant enterococci"[All Fields]) OR "VRE"[All Fields]) AND ("Wild"[All Fields] AND ("bird s"[All Fields] OR "birds"[MeSH Terms] OR "birds"[All Fields])) and google scholar using this search strategy; "Vancomycin resistant *Enterococcus*" IN "wild birds". In addition, titles and references from papers that were included in the search were used as a supplement. To reduce bias, two authors conducted the search.

Inclusion and exclusion criteria for studies

Included were any cross sectional or longitudinal research on wild birds providing information on the occurrence of VRE isolates and the proportion of both VRE and total enterococci strains. Studies that used the standard methodology and were reported in English were also included.

The following type of articles were excluded: case reports, reviews, comments, studies without the required data, articles on antimicrobial susceptibility evaluations besides vancomycin, articles that failed to report enterococcal isolates distinctly, and articles that failed to report the occurrence of VRE in wild bird species.

Data extraction

Studies that matched our criteria for exclusion were noted, and those that matched our criteria for inclusion were evaluated based on their title, abstract, and complete text. From the publications, the first author's name, publication year, the country the research was conducted, the types of wild birds, the number of cases examined, the detection strategy, the sample size, and the occurrence of VRE in wild bird species were all gathered. A third reviewer checked the conclusions after two reviewers had compiled all the data from the included studies. Consensus was reached in order to reconcile the reviewers' disagreements.

Study quality assessment

The quality of the studies included in this review was appraised using the Joanna Briggs Institute (JBI) critical assessment checklist for prevalence data [18] (Supplementary File S2) and this checklist analyzes nine items. Each question is given a yes or no, uncertain or not applicable response. Yes received a mark of 1, while no and uncertain responses received zero points. The average score for each item was then calculated. The quality of research was then rated as being either poor or good depending on whether they had scores below or above the mean [18]. Studies were considered for the analysis if two reviewers could concur on them. On a rating scale

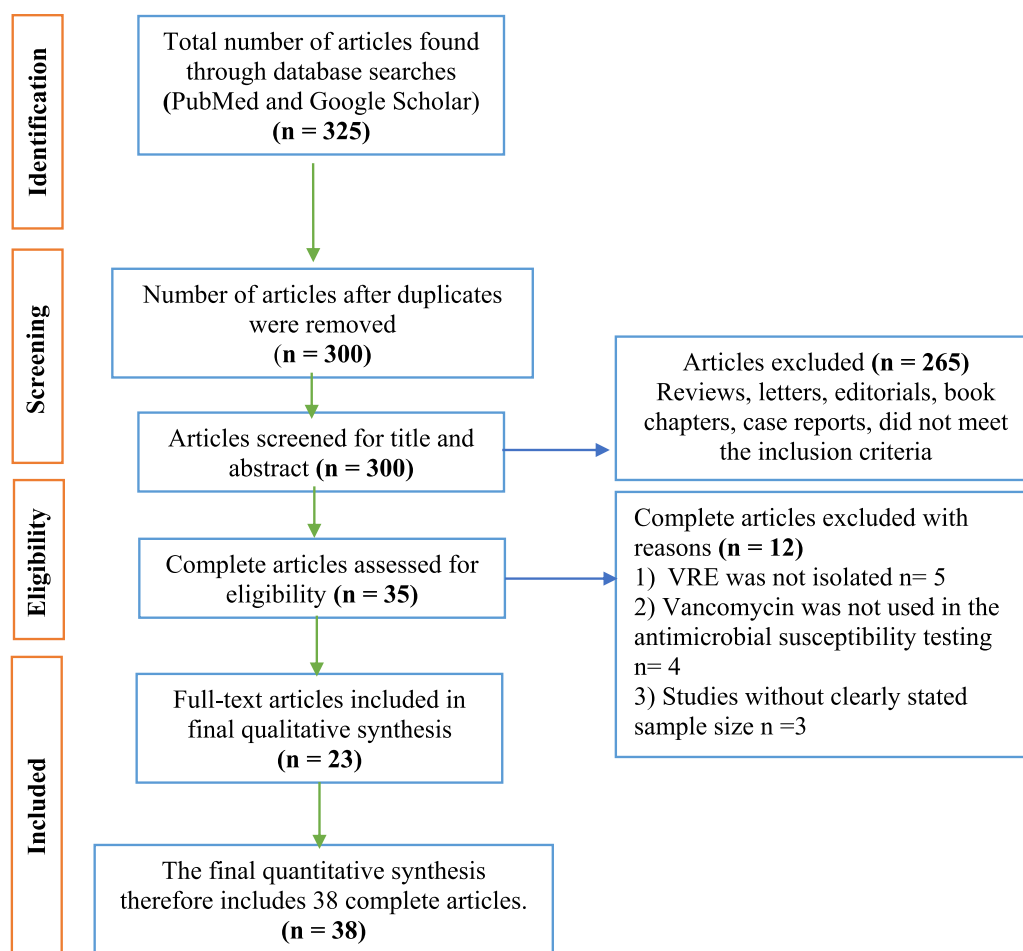


Fig. 1. PRISMA process used to choose the study's eligible articles. Here studies are identified from databases, screened by two reviewers to determine their eligibility before they are added in the final meta-analysis.

Table 1
Characteristics of Included Studies Reporting the Prevalence of VRE in Species of Wild birds.

S/No	Author, Publication Year	Country	Wild bird type	Sample type	Sample Size	Number Positive	Prevalence (%)	Detection method
1	Devriese et al., 1996	Belgium	Pheasants	Fecal	14	2	14.29	Agar dilution
2	Radhouani et al., 2010	Portugal	Sea Gulls	Fecal	57	6	10.53	Agar dilution
3	Radhouani et al., 2010	Portugal	Buzzards	Fecal	33	12	36.36	Agar dilution
4	Goncalves et al., 2010	Portugal	Ostriches	Fecal	54	7	12.96	Disk diffusion
5a	Silva et al., 2011	Portugal	European Robin	Fecal	13	2	15.38	Agar dilution
5b	Silva et al., 2011	Portugal	Common Chaffinch	Fecal	13	3	23.08	Agar dilution
5c	Silva et al., 2011	Portugal	Blackcap	Fecal	13	5	38.46	Agar dilution
5d	Silva et al., 2011	Portugal	Quail	Fecal	13	3	23.08	Agar dilution
6	Ishihara et al., 2013	USA	American Robins	Saliva	16	1	6.25	E test
7	Santos et al., 2013	Portugal	Passerine	Fecal	138	1	0.72	Disk diffusion
8	Oracova et al., 2013	Europe	Rook	Fecal	1073	62	5.78	Broth Microdilution
9	Klibi et al., 2014	Tunisia	Wild birds	Fecal	73	6	8.21	Disk diffusion
10	Oravcova et al., 2014	USA	Crows	Fecal	590	15	2.54	Disk diffusion
11	Oravcova et al., 2014	Canada	Crows	Fecal	100	1	1.00	Disk diffusion
12	Splichalova et al., 2015	Czech Republic	King Fishers	Smear	14	2	14.28	Disk diffusion
13a	Lozano et al., 2016	Spain	White Stork	Swabs	81	40	49.38	Agar dilution
13b	Lozano et al., 2016	Spain	Red Kites	Swabs	59	42	71.19	Agar dilution
13c	Lozano et al., 2016	Spain	Red-Legged Partridge	Swabs	127	10	7.87	Agar dilution
14a	Oravcova et al., 2016	Slovakia	Raven	Fecal	287	4	1.39	Disk diffusion
14b	Oravcova et al., 2016	Slovakia	Rook	Fecal	99	1	1.01	Disk diffusion
15	Roberts et al., 2016	USA	Crows	Fecal	156	38	24.36	Agar dilution
16	Oracova et al., 2017	Australia	Silver Gulls	Fecal	344	2	0.58	Disk diffusion
17	Silva et al., 2017	Portugal	Red-Legged Partridge	Fecal	305	6	1.97	Agar dilution
18a	Ben Yahia et al., 2018	Tunisia	Cattle Egret	Fecal	115	7	6.09	Agar dilution
18b	Ben Yahia et al., 2018	Tunisia	Common Ringed Plover	Fecal	3	1	33.33	Agar dilution
18c	Ben Yahia et al., 2018	Tunisia	White Stork	Fecal	4	1	25.00	Agar dilution
18d	Ben Yahia et al., 2018	Tunisia	Black-Winged Stilt	Fecal	14	1	7.14	Agar dilution
19	Stepien-Pysniak et al., 2018	Poland	Wild birds	Swabs	25	1	4.00	Agar dilution
20a	Kutkowska et al., 2019	Poland	Rooks	Swabs	23	15	65.21	Broth microdilution
20b	Kutkowska et al., 2019	Poland	Wild birds	Swabs	23	8	34.78	Broth microdilution
21a	Smith et al., 2019	Australia	Greater Crested Tern	Swabs	14	2	14.29	Disk diffusion
21b	Smith et al., 2019	Australia	Willie wagtail	Swabs	14	1	7.14	Disk diffusion
21c	Smith et al., 2019	Australia	Varied Honeyeater	Swabs	14	1	7.14	Disk diffusion
21d	Smith et al., 2019	Australia	Sacred Kingfisher	Swabs	14	1	7.14	Disk diffusion
21e	Smith et al., 2019	Australia	Masked booby	Swabs	14	1	7.14	Disk diffusion
21 f	Smith et al., 2019	Australia	Lesser Frigate Bird	Swabs	14	1	7.14	Disk diffusion
22	Cabral et al., 2020	Brazil	Parrot	Swabs	36	2	5.56	Disk diffusion
23	Dolka et al., 2020	Poland	Racing Pigeons	Swabs	145	68	46.90	Disk diffusion

*References for Author publications are listed in [Supplementary File S43.2 The Pooled Prevalence of VRE in Species of Wild Birds](#)

ranging from one to nine, the 23 studies are assessed for quality ([Supplementary File S3](#)).

Data analysis

According to the country where the research was conducted, wild bird type, sources of isolates, method utilized in the detection, subgroup analyses were done to evaluate the prevalence of VRE in various wild bird species.

The DerSimonian and Laird method of meta-analysis was utilized in evaluating the pooled prevalence [18,19]. Heterogeneity was expected because the experiments were conducted in a variety of locations, hence, the adoption of the random effect model.

Bias and heterogeneity

Biases within each study were evaluated using the country the research was conducted, wild bird type, isolation sources, and detection technique. We utilized funnel plots and trim-and-fill plots to search for bias or modest study effects. To examine the heterogeneity of study-level results, Q test was applied.

If the Q/df ratio was below one, non-significant heterogeneity remained tolerated. The amount of disparity in estimating

prevalence was calculated using the inverse variance index (I^2); Low, moderate, and high levels of heterogeneity are indicated by I^2 values of 25%, 50%, and 75%, respectively [20]. Sensitivity analysis, subgroup analysis, and meta regression were utilized in investigating the origins of heterogeneity. OpenMeta Analyst software [21] and Comprehensive meta-analysis version 2 [22] were utilized in conducting the analysis.

Results

325 studies were discovered using database searches, with 25 duplicates being removed. After reviewing the titles and abstracts of the remaining 300 papers, it was discovered that 265 did not match any of the requirements for inclusion. 35 full-text articles were evaluated for eligibility, and 12 were disqualified due to incomplete data and the absence of vancomycin in the antimicrobial susceptibility testing ([Fig. 1](#)). 23 full-text papers in total were utilized in the qualitative analysis ([Fig. 1](#)). To obtain a close approximation of the prevalence of VRE in wild birds, research detailing the prevalence of VRE in multiple species of wild birds were investigated as distinct studies. More than one species of wild bird was shown to have VRE in eight studies ([Table 1](#)). Thirty-eight studies were consequently included in the final analysis.

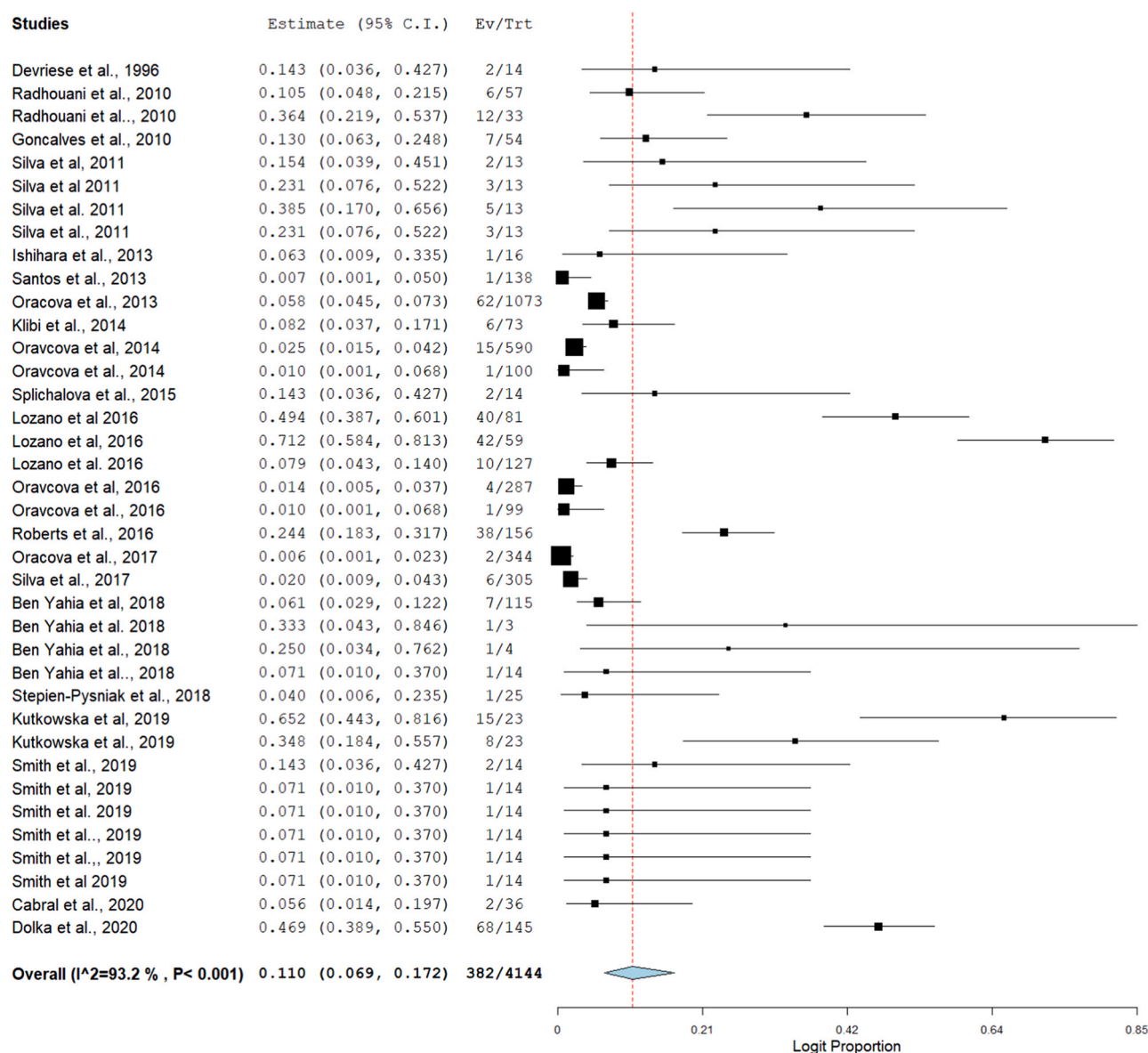


Fig. 2. Forest plot showing the pooled prevalence of VRE in species of wild birds with pooled prevalence at 11.0% (CI: 6.9 – 17.2%) and I^2 at 93.2%.

The pooled prevalence of VRE in species of wild birds was estimated at 11.0% (95% CI; 6.9 – 17.2%; $I^2 = 93.204\%$; $P < 0.001$) (Fig. 2). The variability between the studies was high ($t^2 = 2.156$; $I^2 = 93.204\%$ with $Q = 544.413$, $df = 37$, and $P < 0.001$). No single study had an impact on the variability and aggregate prevalence of VRE in wild bird species as observed in the sensitivity analysis's leave-one-out forest plot (Fig. 3). Moreover, publication bias was detected, as evidenced by the asymmetric funnel plot. (Fig. 4). The Trim-and-Fill approach was then used to include the "missing" studies from the analysis in addition to the funnel plots. The unbiased effect was located by trimming the asymmetric studies, and the trimmed studies were re-inserted. Accordingly, 11 studies were missed and fell at the right side of the pooled estimate (Fig. 5). In the Trim-and-Fill approach, the adjusted estimate of VRE in species of wild birds was 20.48% (95% CI; 13.55 – 29.75%). The Egger's regression $t = 1.1248$ (intercept = -1.1838 ; 95% CI; $-3.32 - 0.95$; $P = 0.2680$) and Begg's rank test ($P = 0.7821$) did not suggest significant publication bias.

Subgroup meta-analysis

To identify the possible sources of heterogeneity among studies, as substantial heterogeneity was observed, subgroup analysis was done using the species of wild birds reported, the country the research was conducted, wild bird type, isolation sources, and detection technique. The outcome of subgroup meta-analysis across the country of the study showed that there was generally significant variation in researches reporting the prevalence of VRE in different species of wild birds. Majority of the studies were reported in Portugal ($n=9$) and Australia ($n=7$) with a pooled prevalence of 12.7% and 5.2% respectively. The Heterogeneity was moderate in studies reported from Australia ($I^2 = 51.99\%$) and high in Portugal ($I^2 = 85.53\%$). However, Spain with 3 studies reported had the highest pooled prevalence (Table 2).

The forest plot for this subgroup meta-analysis is shown in Fig. 6 and a world map showing the distribution of the studies reporting the VRE in wild bird species is shown in Fig. 7.

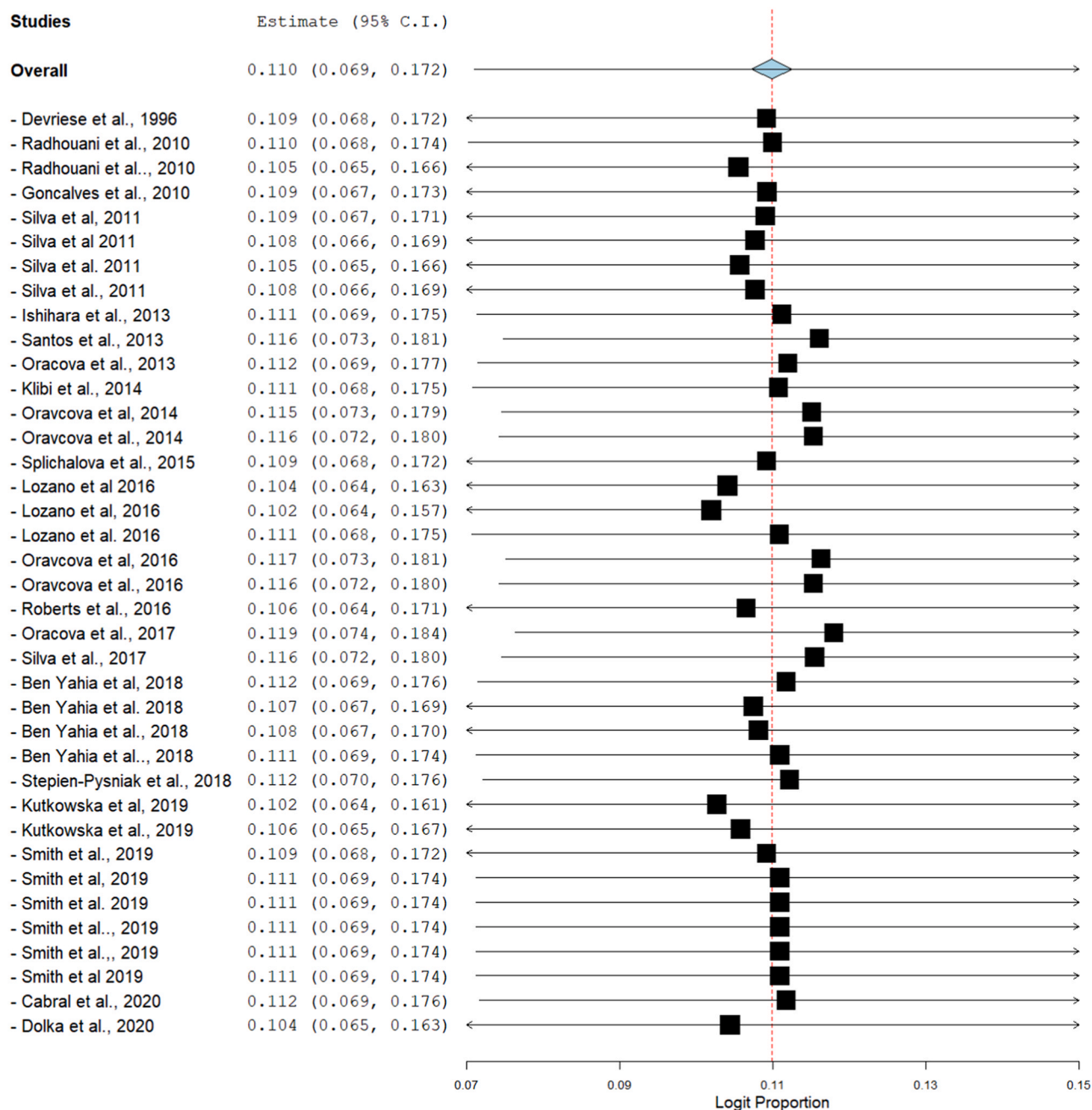


Fig. 3. Sensitivity analysis of VRE in wild birds using the Leave-one-out forest plot estimate. Here, the estimate remains the same as no single study affected it.

The bulk of the studies ($n = 3$) documented the prevalence of VRE, according to the results of subgroup meta-analysis across species of wild birds. Crows and Rooks have a pooled prevalence of 4.8% and 3.2% respectively. Some studies failed to report the species of wild birds where VRE was detected, therefore we categorized them as 'Wild Birds'. Three studies were reported from this category with a pooled prevalence of 12.7% (Table 3). The lowest heterogeneity was observed in studies that detected VRE in White Stork ($n=2$, $I^2 = 0.00\%$).

Also, the forest plot of subgroup meta-analysis across species of wild birds is seen in Fig. 8.

Furthermore, the result of subgroup meta-analysis according to isolate sources revealed that most of the studies ($n=22$) had their VRE isolated from fecal sample with a prevalence of 7.7% and a high

heterogeneity ($I^2 = 88.76\%$) (Table 4). This was followed 14 studies reporting the isolation of VRE from swabs with a prevalence of 20.95% and also a high heterogeneity ($I^2 = 88.67\%$) (Table 4). The forest plot is also shown in Fig. 9.

Agar dilution and disc diffusion were both used in 17 studies, with respective prevalence of 5.0% and 18.4%, according to the results of subgroup meta-analysis based on the detection method. (Table 5). Only one study utilized E Test as a detection method (6.3% and CI of 0.9 - 33.5%) (Table 5) and Fig. 10 shows the forest plot.

Meta-regression

A univariate analysis was done using methods of moment as computational option for each variable included and all of the

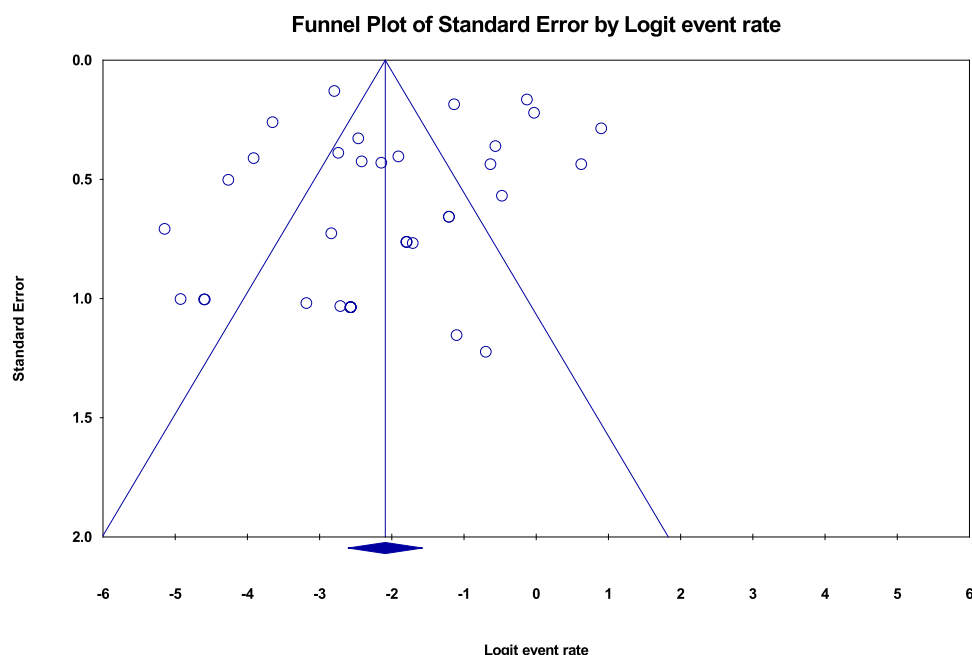


Fig. 4. Funnel plot showing publication bias in studies reporting the prevalence of VRE in species of wild birds. Studies on the left side are fewer than those on the right and thus asymmetrical. The funnel plot is used for the visualization of bias.

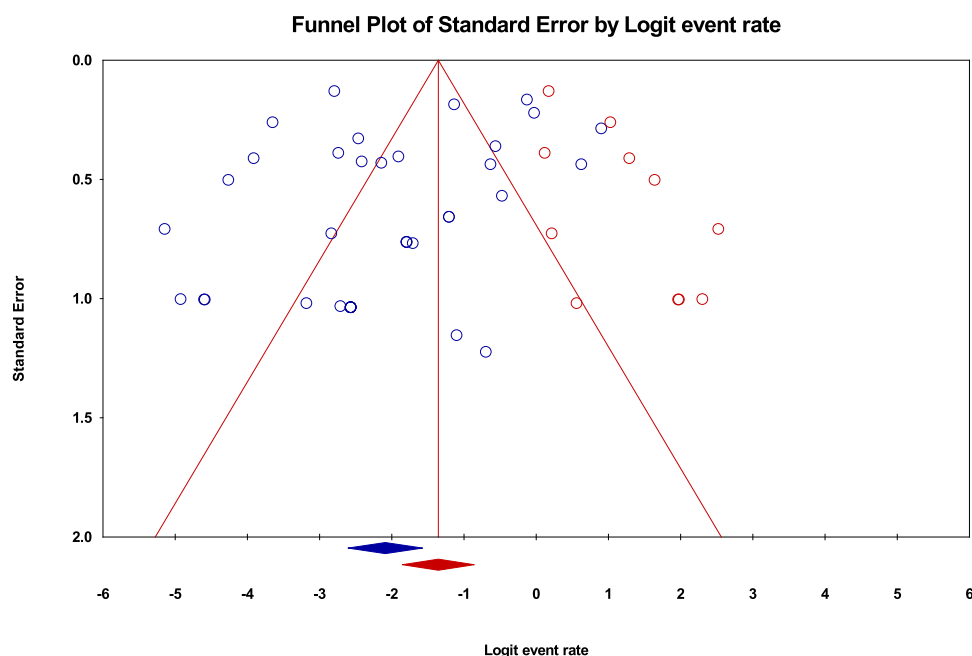


Fig. 5. Trim-and- Fill approach reporting the prevalence of VRE in species of wild birds.

variables played a role in the study's observed heterogeneity (Table 6).

Discussion

This is the first study to evaluate the status of VRE in various wild bird species using meta-analysis and a systematic review. In this study, the prevalence obtained is by analyzing data from published

articles on the occurrence of VRE in wild bird species. A meta-analysis was conducted on 38 studies. The literature evaluation includes VRE reports from several countries, multiple kinds of wild birds, varied isolate sources, and various techniques for VRE detection, therefore it was to be expected that the literature would be diverse. A random effect size model was consequently applied. Due to publication bias and small-study effects, this study's substantial heterogeneity may have been caused by the fact that smaller studies

Table 2

Analysis of subgroups to compare VRE in various wild bird species across countries.

Country of Study	Number of studies	Prevalence (%)	95% CI	I^2 (%)	Q	Heterogeneity test	
						DF	P
Belgium	1	14.3	3.6 – 42.7	-	-	-	-
Portugal	9	12.7	5.6 – 26.4	85.53	55.290	8	< 0.001
USA	3	7.8	1.1 – 39.0	96.76	61.692	2	< 0.001
Europe	1	5.8	4.5 – 7.3	-	-	-	-
Tunisia	5	8.1	5.0 – 12.9	0.00	3.975	4	0.409
Canada	1	1.0	0.1 – 6.8	-	-	-	-
Czech Republic	1	14.3	3.6 – 42.7	-	-	-	-
Spain	3	37.3	9.4 – 77.3	96.78	62.055	2	< 0.001
Slovakia	2	1.3	0.5 – 3.1	0.00	0.084	1	0.772
Australia	7	5.2	2.0 – 13.0	51.99	12.498	6	0.052
Poland	4	39.3	21.4 – 60.6	77.24	13.181	3	0.004
Brazil	1	5.6	1.4 – 19.7	-	-	-	-

CI: Confidence Interval, I^2 : Heterogeneity, Q: heterogeneity chi-square, DF: Degree of freedom, P: P-value

occasionally exhibit unique and frequently bigger treatment effects than larger ones.

In meta-analysis, determining publication bias is crucial. This is because not all findings are released, especially the negative findings, or findings that will evoke a minimal degree of interest. In contrast to trials with more modest treatment advantages, studies with reasonably high treatment outcome have a better probability of getting submitted and/or published. The significant amount of variability found in this meta-analysis suggests that factors other than chance compensated for the observed variability. The observed heterogeneity was caused by all of the factors that were looked at in these study.

As this is the first study to evaluate the status of VRE in species of wild birds using meta-analysis, there were no prior studies for comparison. However, meta-analysis reporting the prevalence of VRE in companion animals have been reported with a prevalence of 14.6% [23] respectively.

Although wildlife is not typically exposed to clinical antibacterial medications, they can acquire resistant bacterial strains through interaction with a contaminated environment, with feces-polluted water playing a crucial role. Humans and animals undergoing antibiotic therapy have their intestinal microbes constantly released into the environment via wastewater or dung. Sewage and runoff from manure-fertilized fields enter waterways, allowing fecal bacteria to be transferred across vast distances. Furthermore, migrating birds that come into touch with antibiotic-resistant bacteria through contaminated water or garbage operate as carriers and play a key role in the spread of resistance. It's also worth noting that many antimicrobials can be excreted in an active form by humans, farm animals, and pets, and thus remain in the environment, supporting the selection leverage of antibiotic-resistant bacteria [24].

Corvids are major reservoirs and carriers of antibiotic-resistant bacterial strains in the environment [25,26]. The rook is a Palearctic omnivore with a wide range. Rooks leave their roosting location in search of food, typically between 0.5–2 km of the breeding population or even 10–40 km of the resting site during winter [24,27]. As such it is currently the commonest species in agro – ecosystems and also has a part in the dispersal of harmful microbes due to its growing number and movement [25].

Crows, on the other hand, have benefited from human activity. They are found all over the world in mostly urban areas. Crows are very engaged in human-affected situations [28,29]. They eat

human trash from garbage dumps to waste water treatment plants, where they drink potentially VRE-contaminated exposed effluent.

Crows examine cow manure and dirt for insects in wetlands and farms, as well as hunting for food in these areas. They have also dominated urban and agricultural avifaunas globally over time [29].

Seventeen studies employed the disc diffusion method which has been described to be functional, with a well-standardized model and can provide specific data that can be simply understood, in addition to being able to select from a range of discs to test [30]. One of the shortcomings of the disc test is that it is not automated or mechanized. E Test was only used in one study to detect VRE. The E Test is a faster and more sensitive method than others for the identification of various VRE species.

Epidemiological surveillance systems have been set up in response to the global public health concern of which antimicrobial resistance (AMR) is at the forefront, with the goal of fostering co-operation aimed at maintaining the health of humans and animals as well as the ecosystem. Wada et al. [31] had reported a 6.6% pooled prevalence of VRE in wildlife in their study, and encourage the establishment of balance between human and animal health. A Global Action Plan on Antimicrobial Resistance has been developed by a number of international organizations, including the Food and Agriculture Organization of the United Nations (FAO), the World Organization for Animal Health (OIE), and others. This plan's actions included learning about AMR through research and surveillance. In order to guarantee that all nations undertake integrated surveillance that will cover the use and consumption of antibiotics in both human and animal populations, the advisory group established guidelines for AMR surveillance. These guidelines will give the public a comprehensive knowledge of how AMR spreads in various contexts and geographical locations. It will make it possible to evaluate the impact of interventions within and between sectors as well as investigate the relationship between AMR and antibiotic usage in a variety of settings, including humans, animals, and the environment. WHO introduced the Global Antimicrobial Resistance Surveillance System (GLASS) in 2015 to close information gaps and direct approaches across the board. GLASS was developed to track the usage of antibiotics, gradually incorporate surveillance data on their use in humans, and comprehend the role of antimicrobial resistance (AMR) in the environment and food chain. It offers a standardized method for gathering, evaluating, interpreting, and

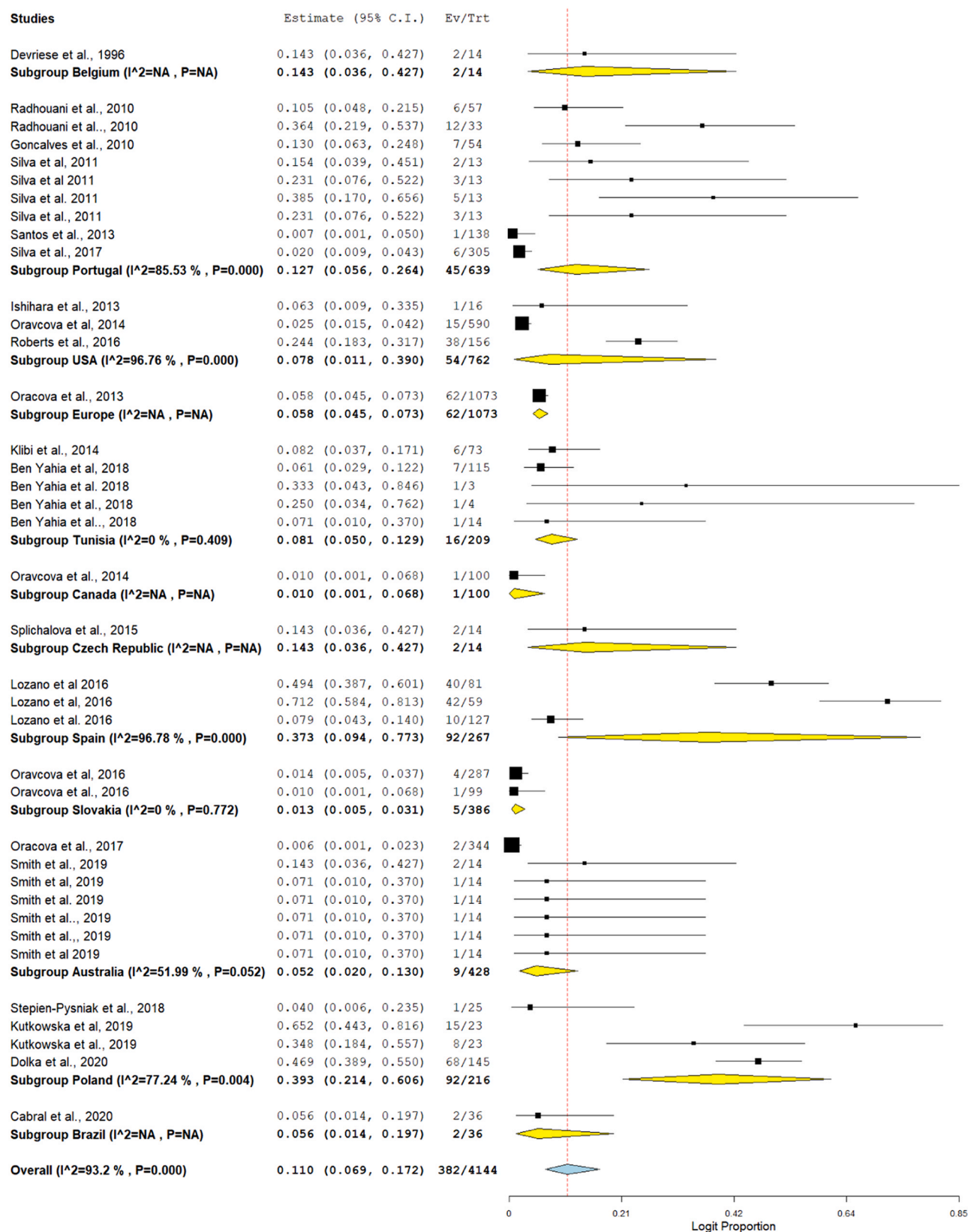


Fig. 6. Forest plot displaying the meta-analysis of the VRE in wild bird species by countries.

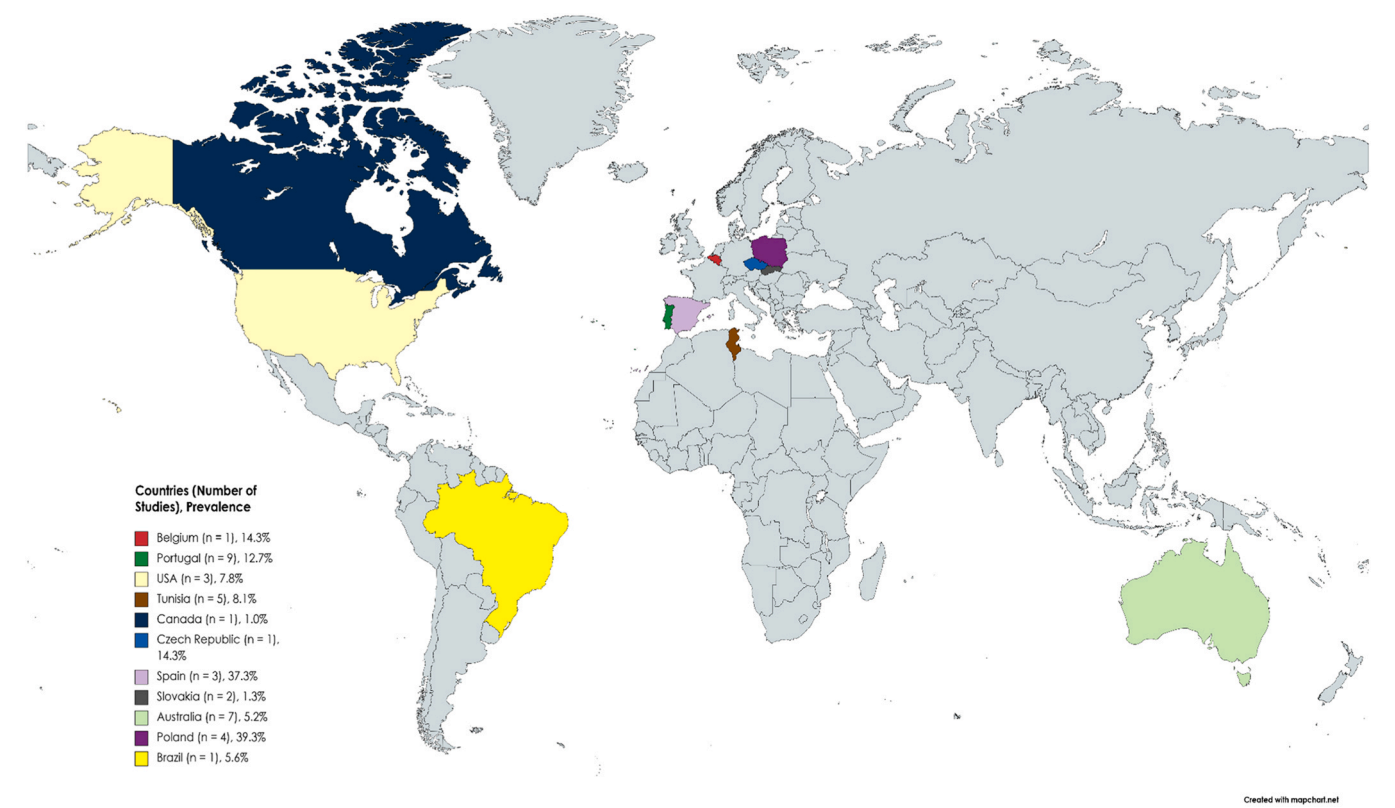


Fig. 7. Map Showing the Global Distribution of VRE in Species of Wild Birds, the Number of Studies and Pooled Prevalence.

Table 3
Analysis of subgroups to compare VRE in various wild bird species.

Bird Species	Number of studies	Prevalence (%)	95% CI	I ² (%)	Q	Heterogeneity test	
						DF	P
Pheasants	1	14.3	3.6 – 42.7	-	-	-	-
Sea Gulls	1	10.5	4.8 – 21.5	-	-	-	-
Buzzards	1	36.4	21.9 – 53.7	-	-	-	-
Ostrich	1	13.0	6.3 – 24.8	-	-	-	-
European Robin	1	15.4	3.9 – 45.1	-	-	-	-
Common Chaffinch	1	23.1	7.6 – 52.2	-	-	-	-
Blackcap	1	38.5	17.0 – 65.6	-	-	-	-
Quail	1	23.1	7.6 – 52.2	-	-	-	-
American Robins	1	6.3	0.9 – 33.5	-	-	-	-
Passerine	1	0.7	0.1 – 5.0	-	-	-	-
Rook	2	3.2	0.6 – 14.9	68.06	3.131	1	0.077
Wild birds	3	12.7	3.2 – 39.2	81.62	10.880	2	0.004
Crows	3	4.8	0.6 – 29.1	97.05	67.824	2	< 0.001
King Fishers	1	14.3	3.6 – 42.7	-	-	-	-
White Stork	2	48.4	38.0 – 59.0	0.00	0.834	1	0.361
Red Kites	1	71.2	58.4 – 81.3	-	-	-	-
Red-Legged Partridge	2	4.1	1.0 – 14.9	86.73	7.538	1	0.006
Raven	1	1.4	0.5 – 3.7	-	-	-	-
Silver Gulls	1	0.6	0.1 – 2.3	-	-	-	-
Cattle Egret	1	6.1	2.9 – 12.2	-	-	-	-
Common Ringed Plover	1	33.3	4.3 – 84.6	-	-	-	-
Black-Winged Stilt	1	7.1	1.0 – 37.0	-	-	-	-
Rooks	1	65.2	44.3 – 81.6	-	-	-	-
Greater Crested Tern	1	14.3	3.6 – 42.7	-	-	-	-
Willie wagtail	1	7.1	1.0 – 37.0	-	-	-	-
Varied Honeyeater	1	7.1	1.0 – 37.0	-	-	-	-
Sacred Kingfisher	1	7.1	1.0 – 37.0	-	-	-	-
Masked booby	1	7.1	1.0 – 37.0	-	-	-	-
Lesser Frigate Bird	1	7.1	1.0 – 37.0	-	-	-	-
Parrot	1	5.6	1.4 – 19.7	-	-	-	-
Racing Pigeons	1	46.9	38.9 – 55.0	-	-	-	-

CI: Confidence Interval, I²: Heterogeneity, Q: heterogeneity chi-square, DF: Degree of freedom, P: P-value

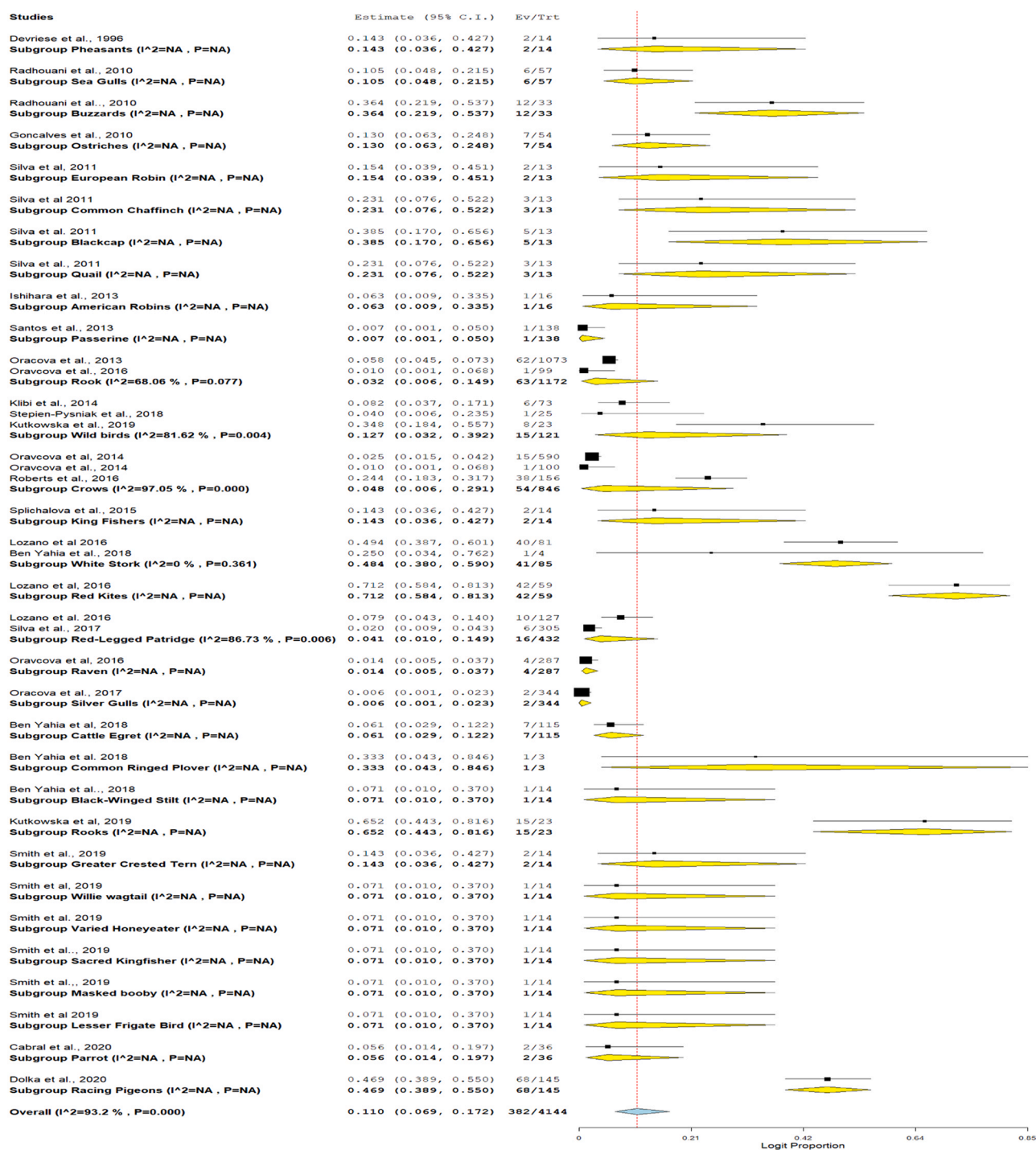


Fig. 8. A forest plot displaying the meta-analysis of VRE by species of wild birds.

Table 4

Analysis of subgroups to compare VRE in various wild bird species across different isolates.

Isolate Source	Number of studies	Prevalence (%)	95% CI	I^2 (%)	Q	Heterogeneity test	
						DF	P
Faecal	22	7.7	4.6 – 12.5	88.76	186.777	21	< 0.001
Saliva	1	6.3	0.9 – 33.5	-	-	-	-
Smear	1	14.3	3.6 – 42.7	-	-	-	-
Swabs	14	20.9	11.6 – 34.8	88.67	114.721	13	< 0.001

CI: Confidence Interval, I^2 : Heterogeneity, Q: heterogeneity chi-square, DF: Degree of freedom, P: P-value

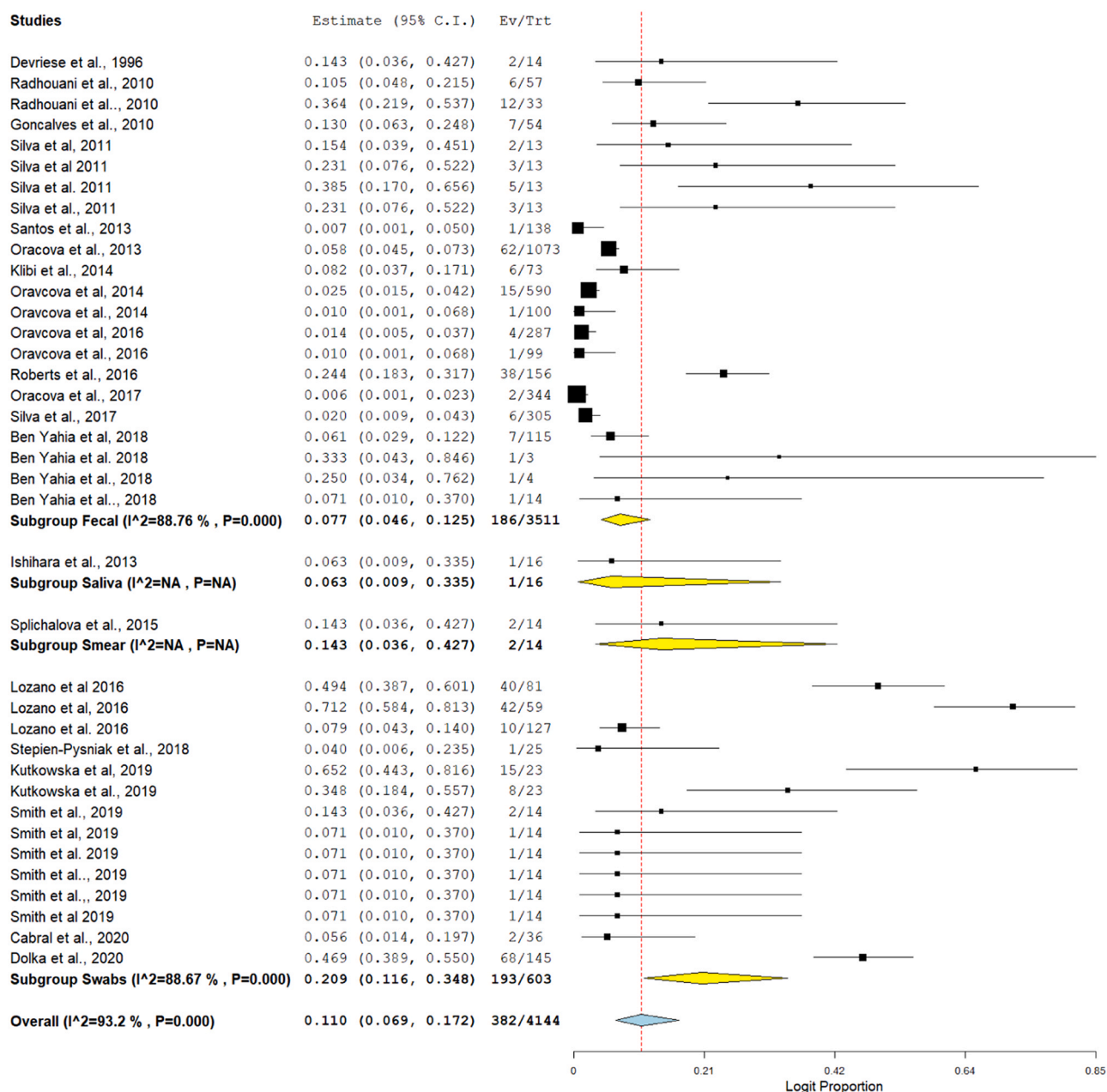


Fig. 9. Forest plot displaying the subgroup analysis by Isolate sources of VRE in species of wild birds.

Table 5

Analysis of subgroups to compare VRE in various wild bird species across detection methods.

Detection method	Number of studies	Prevalence (%)	95% CI	I ² (%)	Q	Heterogeneity test	
						DF	P
Agar Dilution	17	18.4	10.3 – 30.6	90.59	170.092	16	< 0.001
Disk diffusion	17	5.0	2.0 – 12.0	93.02	229.220	16	< 0.001
E Test	1	6.3	0.9 – 33.5	-	-	-	-
Broth Microdilution	3	27.7	3.9 – 78.6	97.26	73.116	2	< 0.001

CI: Confidence Interval, I²: Heterogeneity, Q: heterogeneity chi-square, DF: Degree of freedom, P: P-value

disseminating data by nation, area, and region. It also lets you keep track of the development of new or current national surveillance systems while highlighting the accuracy and representativeness of

the data gathered. In order to direct the development of novel antibiotics, diagnostic instruments, and vaccines, the World Health Organization created a list of priority pathogens in 2017.

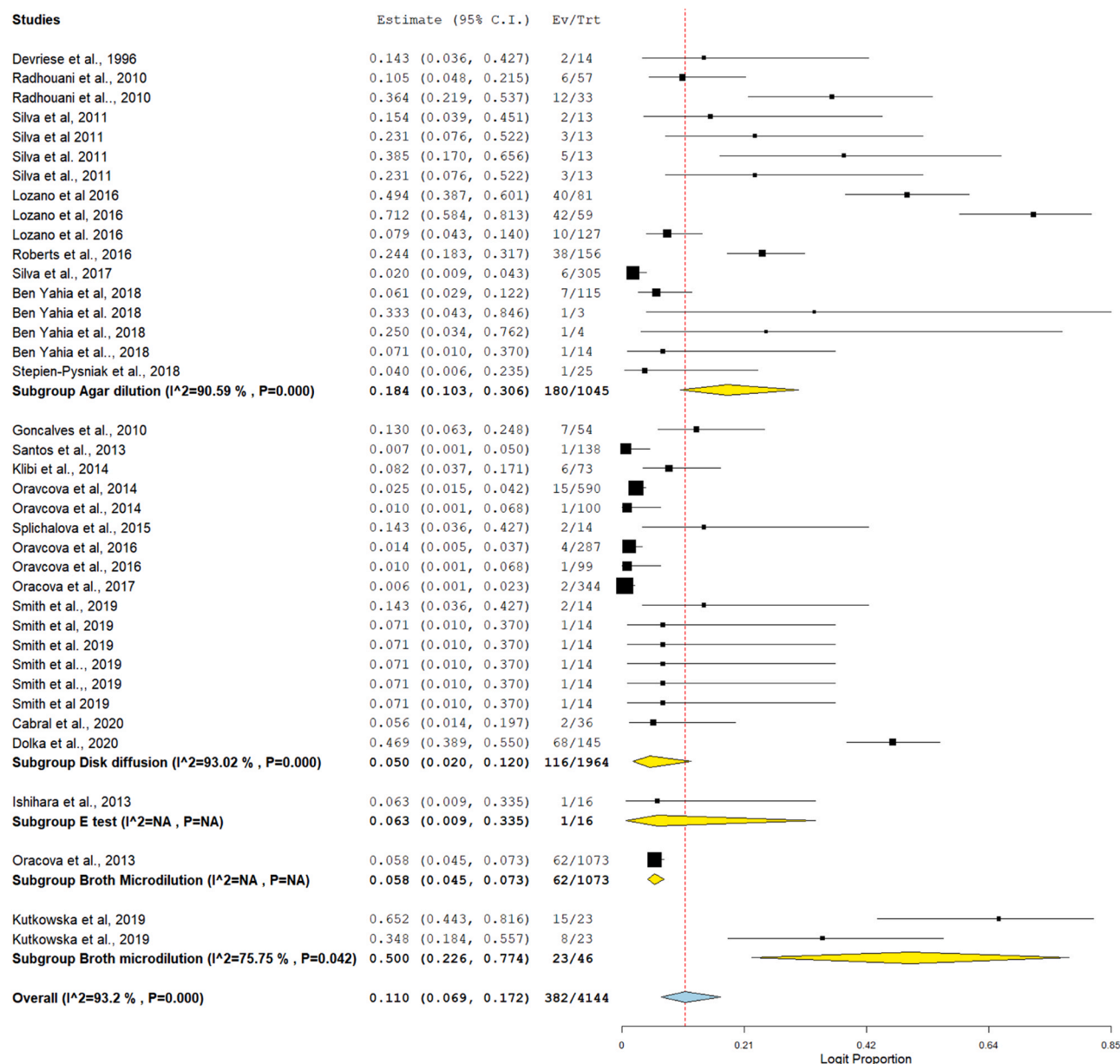


Fig. 10. Forest plot displaying the subgroup meta-analysis based on detection method of VRE in species of wild birds.

Table 6
Univariate meta-regression model of VRE in species of wild birds.

Variable	Q	DF	P Value
Country	30.33155	37	0.77293
Wild Bird type	30.00199	37	0.78594
Isolate Source	49.63058	37	0.08019
Detection method	32.62107	37	0.67445

Conclusion

Without a doubt, birds play a role in the spread of diseases that affect humans. They serve as reservoirs for resistance genes, causing its spread and dispersion across broad areas during migration. A systematic review and meta-analysis of studies that reported the prevalence of VRE in wild bird species yielded a pooled prevalence of 11.0% due to the high volume of samples and a low number of resistant samples. It is difficult to assume that this result accurately represents the true point estimate given the observed reasonably

high heterogeneity. However, this estimate better reflects the prevalence of VRE in wild bird species. The findings in this study are important from public health perspective, especially as we consider this to be the tip of the iceberg. Also, this study further strengthens the case for a One Health approach towards integrated management of infectious diseases.

Tracking antibiotic resistance and pathogenicity in wildlife is a valuable method for learning about the distribution of highly infectious strains and resistance determinants in natural settings. Some limitations encountered in this study was that vancomycin was not used in the AST testing of some studies and some studies did not clearly report their sample sizes. To improve the detection of VRE in different samples, further research should be carried out using molecular tools, that way, the pooled prevalence of VRE in wild bird species globally might be high.

Data availability statement

Data can be found in the manuscripts.

Declaration of competing interest

The authors declare no conflict of interest.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.jiph.2024.04.004](https://doi.org/10.1016/j.jiph.2024.04.004).

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